AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ENGINEERING COIMBATORE

Department of Electrical and Electronics Engineering

19CSE362 · Fundamentals of Soft Computing · Project Review

Explainable ANN–ANFIS Hybrid for Renewable-Aware Task Offloading in Solar-Powered IoT Nodes Using a Digital Twin

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Every minute a solar-powered ESP32 must decide: **run the task itself, or offload it over WiFi to an edge server?** An ANN **predicts** what offloading will cost, and an ANFIS with 36 learned fuzzy rules **decides** and explains why.

How it works: from digital twin to decision

1 · Digital twin: a physics simulator, one step = one minute

- **Solar PV** — 1 W panel; sunlight and clouds change through the day
- **Battery** — 12 Wh; charged by solar, drained by the node
- **ESP32 node** — 240 MHz; one IoT task arrives every minute
- **WiFi / MQTT** — signal, speed and retries change with distance
- **Edge server** — 3 GHz; much slower when it is busy

2 · What the ESP32 knows before deciding

battery level · task size and type · its own local time and energy · last measured signal · last server load

3 · Dataset: 10,080 tasks × 38 features

each task labelled with the cheaper of the two exact costs (the oracle): 54 % local, 46 % offload

trains the ANN and ANFIS (teacher)

4 · ANN predictor

6 → 24 → 16 → 2 neurons; predicts the offload time and energy

5 · ANFIS · 36 rules

4 inputs, 5 layers; fuzzy rules learned from data give an offload score

6 · Decision

score above 0.5 → OFFLOAD to the edge, otherwise run LOCAL

closed loop: the chosen option's energy is drawn from the battery

7 · Evaluation

closed loop on 11 unseen weeks against Fuzzy-only and ANN-only, Wilcoxon test

8 · Explainable dashboard

shows every decision and the rules that fired

96.8 %	**17 J**	**36**	**20 / 20**
agreement with the best possible decision, 11 unseen weeks	less task energy per week than the ANN alone (p ≈ 0.001)	learned fuzzy rules that explain every decision	engineering checks pass on the digital twin